A Regulatory Feasibility Evaluation Framework to Support Practical Medical Device Research Projects
연구과제 기반 실질적 의료기기 지원을 위한 규제 실현가능성 평가체계 제안

GPT-5.2

Abstract

Korea has made substantial national investments in science and technology research, with total government research and development (R&D) expenditure reaching tens of trillions of Korean won annually. In parallel, policy-level programs supporting medical device regulatory approval and commercialization have been institutionalized, aiming to bridge the gap between research outputs and clinical market entry. Nevertheless, a significant proportion of medical device research projects still fail to translate into authorized products, suggesting that regulatory feasibility is insufficiently evaluated during early research planning stages. In this study, we propose a regulatory feasibility scoring framework that systematically assesses intended use, approval classification, regulatory pathway complexity, and clinical and performance evidence requirements. Applying the framework to representative medical device and in vitro diagnostic use cases, we demonstrate how early-stage feasibility assessment can improve alignment between national R&D investment, regulatory requirements, and realistic market entry timelines.

1 Introduction

Korea's national investment in science and technology research is substantial and continues to grow. According to government budget disclosures, the Ministry of Science and ICT (MSIT) alone was allocated approximately **23.7 trillion KRW** in total budget for fiscal year 2026, including **11.8 trillion KRW dedicated to R&D**, contributing to an overall national R&D expenditure of approximately **35.5 trillion KRW**. These figures indicate that even marginal inefficiencies in the translation of research outcomes to practical applications may result in significant systemic loss of public resources.

Within this broader R&D ecosystem, medical devices represent a strategically important sector due to their direct linkage between technological innovation, regulatory authorization, and clinical application. Recognizing persistent translational barriers, the Korean government has established explicit **policy-level support mechanisms for medical device licensing and market entry**. Representative examples include the *Pan-ministerial Medical Device R&D Program (2020–2025)*, a large-scale initiative with a total budget of approximately **1.2 trillion KRW**, designed to support the full development lifecycle from research and development to regulatory approval and commercialization. In addition, targeted regulatory support programs, such as international certification and approval assistance initiatives, have been implemented with dedicated multi-year funding to facilitate global market access.

Despite the existence of these policy instruments, the success rate of medical device authorization and commercialization remains uneven. This discrepancy suggests that policy support alone is insufficient if regulatory requirements are not systematically integrated into early research design. In particular, research proposals often emphasize technological novelty or scientific contribution without adequately accounting for intended use definition, approval classification, regulatory pathway complexity, or evidence generation requirements. As a result, downstream regulatory obstacles emerge late in development, increasing cost, delay, and project attrition.

Accordingly, this study argues that **explicit evaluation of regulatory feasibility should be treated as a core research planning activity**, rather than a post hoc compliance exercise. To address this need, we propose a structured scoring framework that enables comparative assessment of regulatory feasibility across different medical device and in vitro diagnostic intended uses. By quantifying regulatory burden and approval complexity at an early stage, the framework aims to support more realistic research planning, improve alignment with existing licensing support policies, and ultimately enhance the efficiency of national R&D investmen

2 Related Work and Regulatory Basis

2.1 Regulatory Framework for Medical Device Licensing in Korea

In Korea, the regulatory approval of medical devices is governed by the *Regulations on Medical Device Licensing, Reporting, and Review* administered by the Ministry of Food and Drug Safety (MFDS†). Unlike purely technology-driven evaluation frameworks, the Korean regulatory system explicitly anchors approval decisions in **intended use**, **degree of modification**, and **evidence generation requirements**, which are codified in legally binding annex tables.

Annex Table 5†† of the regulations establishes the fundamental logic for determining product equivalence by evaluating whether a device's intended use, principle of operation, and key technological characteristics are comparable to those of an already approved product. Under this framework, devices are classified as *equivalent products*, *modified products*, or *new products*, and this classification directly determines the regulatory pathway and review burden. Importantly, equivalence is assessed not only on material or structural similarity but also on whether the intended clinical use remains unchanged, highlighting the central regulatory role of intended use definition.

2.2 Submission Scope and Clinical Evidence Triggers

Annex Table 7††† specifies the scope of technical documentation required for licensing, reporting, and review according to the product classification outcome. The table delineates which elements—such as performance testing, safety data, biocompatibility evaluation, and clinical evidence—must be submitted for each category. Notably, even products classified as modified may be required to submit clinical data when performance or safety cannot be sufficiently demonstrated through non-clinical testing alone.

This regulatory structure implies that clinical evidence requirements are not binary but conditional, depending on the interaction between intended use, degree of technological modification, and residual risk. As a result, regulatory burden can increase substantially even for devices that appear technologically incremental if the intended use is broadened or reinterpreted.

2.3 Stage-Based Approval and Review Process

Annex Table 15†††† further formalizes the approval process into a stage-based structure, typically progressing from design and development review, through performance and safety verification, to clinical evaluation where applicable. This staged approach provides a regulatory roadmap that implicitly defines expected timelines, decision points, and potential escalation triggers, particularly when a device advances from non-clinical to clinical evaluation phases.

The existence of this staged structure demonstrates that regulatory approval is not a single event but a cumulative process, where early design and intended use decisions exert long-term influence on approval complexity and duration. Consequently, failure to account for these stages during early research planning may result in downstream regulatory delays or the need for unanticipated evidence generation.

2.4 Public Approval Data as Empirical Regulatory Evidence

In addition to formal regulations, the **Medical Device Safety Information Disclosure Database (의료기기 안심책방*)** operated by MFDS provides publicly accessible records of approved devices, including product classification, intended use descriptions, and approval status.

---

†식품의약품안전처고시 제2026-6호 의료기기 허가・신고・심사 등에 관한 규정

††식품의약품안전처고시 제2026-6호 의료기기 허가・신고・심사 등에 관한 규정 [별표 5] 동등제품 판단기준(제23조 관련)

†††식품의약품안전처고시 제2026-6호 의료기기 허가・신고・심사 등에 관한 규정 [별표 7] 기술문서 등 제출 자료의 범위(제28조 관련)

††††식품의약품안전처고시 제2026-6호 의료기기 허가・신고・심사 등에 관한 규정[별표 15] 단계별 제출자료(제51조제1항 관련)

*https://emedi.mfds.go.kr/search/data/MNU20237

This database serves as an empirical reference for understanding how regulatory principles outlined in Annex Tables 5, 7, and 17 are applied in practice.

Analysis of disclosed approval information reveals consistent patterns in how intended use wording influences classification outcomes and submission requirements, reinforcing the regulatory logic articulated in the formal annex tables. As such, the database functions as an important complementary resource for regulatory feasibility assessment and comparative analysis.

2.5 Implications for Regulatory Feasibility Evaluation

Taken together, Annex Tables 5, 7[†††], and 15[††††] and the public approval disclosure database demonstrate that Korea already possesses a **highly structured, rule-based regulatory framework** for medical device approval. However, these regulatory criteria are typically applied *after* research directions and product concepts have been fixed. This temporal separation limits their utility as proactive decision-making tools during early research planning.

Therefore, while policy and regulatory support for medical device licensing clearly exists, the literature and regulatory documents collectively indicate the need for **systematic evaluation of licensing feasibility at the research design stage**. Explicitly translating regulatory rules into evaluative frameworks can help align research objectives with approval realities, reduce late-stage attrition, and improve the efficiency of publicly funded medical device research.

3 Methods

3.1 Conceptual Translation of Regulatory Logic into Scoring Rules

The regulatory feasibility scoring framework proposed in this study was developed by explicitly translating the Korean medical device regulatory logic into quantitative evaluation rules. Rather than introducing new regulatory criteria, the framework is grounded entirely in existing legal and administrative structures defined in the Regulations on Medical Device Licensing, Reporting, and Review, particularly Annex Tables 5[††], 7[†††], and 15[††††], and empirical approval patterns observed in the Medical Device Safety Information Disclosure Database.

Annex Table 5[††] establishes intended use equivalence as the primary determinant of regulatory classification, distinguishing equivalent products, modified products, and new products. Annex Table 7[†††] specifies the scope of technical documentation required for each classification, including conditions under which clinical evidence is triggered. Annex Table 15further structures the approval process into sequential stages, implicitly reflecting increasing regulatory burden and expected development timelines. Together, these regulatory elements form a decision logic that can be systematically mapped to evaluable dimensions.

3.2 Definition of Evaluation Dimensions

Based on the regulatory structure, four core dimensions were defined to represent regulatory feasibility:

Clarity of Intended Use Definition

Approval Classification Difficulty

Regulatory Pathway and Submission Burden

Clinical and Performance Evidence Requirements

Each dimension reflects a distinct regulatory decision layer and is directly traceable to a specific annex table or regulatory practice.

3.3 Scoring Rule Design

Each evaluation dimension was assigned a maximum score of 25 points, resulting in a total feasibility score ranging from 0 to 100. Equal weighting was applied to avoid implicit prioritization of any single regulatory aspect and to maintain interpretability.

3.3.1 Intended Use Clarity (0–25)

Scoring for intended use clarity was revised to be based on the degree to which the stated intended use could be **directly and unambiguously mapped to an existing product category and risk class explicitly listed in the Korean medical device classification tables**, rather than on equivalence determination criteria.

Specifically, intended uses were evaluated according to their alignment with predefined device categories and classes as specified in **Annex 1**‡ **of the Regulation on Medical Device Classification** and, where applicable, the **Annex**‡‡ **on In Vitro Diagnostic Medical Device Classification**. Intended uses that corresponded to a **single, clearly defined product code and risk class** without requiring interpretative judgment received higher scores. In contrast, intended uses spanning multiple anatomical sites, therapeutic purposes, or biological interactions—where classification ambiguity or class escalation was likely—were assigned lower scores due to increased regulatory uncertainty.

This revision reflects a classification-first regulatory perspective, emphasizing **predictability of regulatory categorization** as a primary determinant of feasibility, independent of claims of equivalence.

3.3.2 Approval Classification Difficulty (0–25)

Approval classification scores were derived from Annex Table 5†† equivalence logic. Products reasonably expected to qualify as equivalent devices were assigned the highest scores, while those likely to be classified as modified products received intermediate scores. Intended uses that implied new clinical functions or therapeutic effects, thereby increasing the likelihood of classification as new products, were assigned the lowest scores.

3.3.3 Regulatory Pathway and Submission Burden (0–25)

This dimension reflects the documentation scope specified in Annex Table 7†††. Scores were inversely proportional to the expected volume and complexity of required submissions. Products requiring only basic technical documentation and non-clinical performance testing received higher scores, whereas those requiring extensive safety, biocompatibility, or system-level validation received lower scores.

3.3.4 Clinical and Performance Evidence Requirements (0–25)

Clinical and performance evidence scoring was informed by Annex Table 7††† triggers and the stage-based escalation described in Annex Table 15††††. Products for which clinical evidence could be waived or replaced by analytical or performance validation were assigned higher scores. Conversely, intended uses likely to require clinical investigation or confirmatory clinical data were assigned lower scores due to increased development time and uncertainty.

3.4 Output Interpretation

Large language models (LLMs) were used as auxiliary tools to assist in interpreting regulatory text, standardizing intended use descriptions, and checking internal consistency across scoring dimensions. AI was not used to generate regulatory decisions but to support reproducible application of predefined scoring rules across multiple use cases. This approach enabled scalable analysis while preserving regulatory traceability and human interpretability.

The final regulatory feasibility score represents a comparative indicator of expected approval complexity and market entry readiness. Higher scores indicate clearer regulatory pathways and lower evidence burdens, while lower scores reflect increased uncertainty, extended timelines, and higher licensing risk. The framework is designed for early-stage research planning and comparative assessment, not as a substitute for formal regulatory consultation.

‡식품의약품안전처고시 제2025-23호 의료기기 품목 및 품목별 등급에 관한 규정 [별표 1] 의료기기 품목 및 품목별 등급

‡‡식품의약품안전처고시 제2025-95호 체외진단의료기기 품목 및 품목별 등급에 관한 규정 [별표]

체외진단의료기기 품목 및 품목별 등급

4 Results

4.1 Overview of Regulatory Feasibility Assessment

The proposed regulatory feasibility scoring framework was applied to three representative intended uses spanning low-risk medical devices, in vitro diagnostic reagents, and implantable biomaterials. These use cases were selected to reflect distinct regulatory profiles under the Korean medical device approval system and to evaluate the framework's ability to differentiate feasibility across heterogeneous product categories.

For each intended use, scores were calculated across four regulatory dimensions—intended use clarity, approval classification difficulty, regulatory pathway and submission burden, and clinical and performance evidence requirements—resulting in a total feasibility score ranging from 0 to 100.

4.2 Case 1: Sterilized Needle Used to Inject Drugs or Aspirate Body Fluids

Total score: 98 / 100

Regulatory interpretation summary

This product represents a typical **single-use invasive medical device** with transient tissue contact. The intended use is highly standardized, and numerous predicate products have already been approved under the same item classification. Clinical investigation is not required, and approval relies primarily on conformity to established standards.

A. Intended use clarity (25 / 25)

The intended use—injecting drugs or aspirating body fluids—is explicit, function-based, and universally understood. The target, method of use, and anatomical interaction are fully standardized, allowing direct mapping to a single item classification without ambiguity.

B. Approval classification difficulty (25 / 25)

The product is highly likely to be classified as an equivalent device due to identical intended use, principle of operation, and material characteristics compared to existing approved products.

C. Regulatory pathway and submission burden (23 / 25)

Submission requirements are limited to basic technical documentation, sterilization validation, and biocompatibility testing. Most advanced documentation elements are exempt.

D. Clinical and performance evidence risk (25 / 25)

Clinical studies are not required. Compliance can be demonstrated through standardized physical and performance testing alone.

4.3 Case 2: NGS Reagent for Detecting Histocompatibility Antigen (HLA) Class I

Total score: 80 / 100

Regulatory interpretation summary

This product is classified as an **in vitro diagnostic medical device reagent**. While the analytical target and method are clearly defined, regulatory interpretation depends on diagnostic purpose and clinical context, introducing moderate classification complexity.

A. Intended use clarity (20 / 25)

The detection of HLA Class I antigens using NGS is analytically precise. However, regulatory clarity is slightly reduced because classification may vary depending on whether the test is positioned for transplantation compatibility assessment, supplementary diagnosis, or other clinical contexts.

B. Approval classification difficulty (20 / 25)

The product generally aligns with existing IVD item categories, but may fall at the boundary between equivalent and modified devices depending on panel composition, interpretation claims, and clinical application.

C. Regulatory pathway and submission burden (18 / 25)

Required documentation focuses on analytical performance (accuracy, precision, sensitivity, specificity, and interference). The burden is moderate but predictable within the IVD regulatory framework.

D. Clinical and performance evidence risk (22 / 25)

As a qualitative or supportive diagnostic tool, clinical investigation requirements are limited. Performance validation is largely analytical rather than clinical.

4.4 Case 3: Collagen-Containing Medical Device for Skin Regeneration

Total score: 44 / 100

Regulatory interpretation summary

This product exhibits the highest regulatory uncertainty. The intended use implies **therapeutic and regenerative effects**, and the device involves prolonged interaction with biological tissue. Classification often requires case-by-case interpretation and extensive supporting evidence.

A. Intended use clarity (12 / 25)

"Skin regeneration" represents a broad and outcome-oriented claim rather than a narrowly defined function. The intended use does not map directly to a single item classification and may be interpreted across multiple categories (implantable device, tissue-interacting material, regenerative scaffold).

B. Approval classification difficulty (10 / 25)

Due to biological interaction and regenerative implications, the product is likely to be classified as a modified or new device, rather than an equivalent product.

C. Regulatory pathway and submission burden (12 / 25)

Extensive documentation is required, including biocompatibility, degradation behavior, material characterization, and potentially long-term safety data.

D. Clinical and performance evidence risk (10 / 25)

Given invasive application and regenerative claims, escalation to clinical evaluation is highly likely, significantly increasing development time and regulatory risk.

4.5 Implications of Results

The results of this study demonstrate that regulatory feasibility is not a uniform property of medical device development, but a function of how clearly the intended use is defined and how predictably it aligns with existing regulatory classifications. Importantly, not all regulatory approvals require clinical trials, and premature assumptions regarding clinical evidence often lead to inefficient allocation of research and development resources.

Recent policy developments in the United States illustrate how field-specific artificial intelligence tools are increasingly used to support regulatory and administrative decision-making. For example, the U.S. Food and Drug Administration (FDA) has introduced Elsa, an agency-wide generative AI tool designed to improve the efficiency of internal regulatory tasks such as clinical protocol review and performance optimization. Similarly, the Centers for Medicare & Medicaid Services (CMS) has deployed the WISeR model to identify and manage wasteful or inappropriate services through domain-specific AI applied to predefined policy objectives.

These examples highlight a common principle: AI systems deliver the greatest value when applied within clearly bounded regulatory contexts, where decision criteria and objectives are well defined. In the medical device domain, the intended use plays a role analogous to these policy objectives. Whether a temperature-controlled device is intended to store sandwiches or blood fundamentally alters the regulatory framework, evidence requirements, and scale of resources necessary to enter the regulated market. The findings of this study suggest that feasibility scoring based on intended use clarity can function as an early indicator of such downstream resource implications..

5 Discussion

An important implication of this study is that **not all medical device research projects require clinical trials**, nor should clinical validation be treated as a uniform or default endpoint in research funding and evaluation. In regulated development, the necessity and scale of clinical evidence are determined primarily by the **intended use and regulatory strategy**, rather than by technological novelty alone.

When approval support is an explicit objective of a research project, it becomes critical to distinguish whether the goal is to incrementally improve an existing product already in clinical use, or to introduce a device with a new regulatory positioning. These pathways entail markedly different regulatory burdens and **substantially different scales of clinical support**. In practice, approval-related support may range from projects requiring no clinical involvement and relying on equivalence or standards-based conformity, to projects necessitating limited clinical performance verification, and ultimately to projects that demand full-scale clinical investigations due to higher risk classification or novel intended use.

The results of this study suggest that failing to make such distinctions at the project evaluation stage can lead to inefficient allocation of regulatory and clinical resources. Overestimating clinical needs may divert funding toward unnecessary trials, while underestimating them may leave genuinely high-risk or innovative projects under-supported. By contrast, feasibility scoring based on intended use clarity enables a more **resource-rational regulatory strategy**, allowing funding bodies to align approval support and clinical investment with the actual regulatory requirements of each project.

In this sense, regulatory feasibility assessment functions not as a gatekeeping mechanism, but as an early decision-support tool for estimating the **appropriate scale of approval-related and clinical support**. This aligns with recent policy trends toward targeted, domain-specific decision support systems, where the objective is not to standardize outcomes, but to improve transparency and predictability in complex regulatory environments.

6 Limitations and Future Work

This study has several limitations. First, the feasibility scoring framework is based on current regulatory classification tables and publicly disclosed approval data, which may evolve over time as regulations are revised or new device categories are introduced. Second, the scoring system emphasizes intended use clarity and classification predictability and does not explicitly model post-market obligations, reimbursement considerations, or international regulatory divergence.

Future work may extend the framework by incorporating dynamic regulatory updates, cross-jurisdictional comparisons, and post-market surveillance requirements. In addition, integration of domain-specific AI tools—similar in concept to FDA's Elsa or CMS's WISeR—could enable semi-automated feasibility assessment at scale, supporting portfolio-level analysis for funding agencies and research institutions.

Ultimately, the framework presented here is not intended to replace regulatory review or expert consultation, but to provide a structured, transparent mechanism for anticipating regulatory feasibility and resource requirements at the earliest stages of medical device research.

Acknowledgments and Disclosure of Funding

The author would like to acknowledge the use of artificial intelligence–based tools in supporting the research process, including regulatory document interpretation, analytical structuring, and manuscript drafting assistance. These tools were used to enhance clarity, consistency, and efficiency, while all analytical decisions, interpretations, and conclusions remain the sole responsibility of the author.

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. No external financial support was provided for the conduct of this study.

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The use of AI tools did not involve proprietary data, confidential information, or restricted regulatory materials.

References

[1] Ministry of Food and Drug Safety (MFDS) (2025). *Regulations on Medical Device Items and Grades by Item*. MFDS Notification No. 2025-23. Appendix 1: Medical Device Items and Grades by Item. Cheongju, Republic of Korea.

[2] Ministry of Food and Drug Safety (MFDS) (2025). *Regulations on In Vitro Diagnostic Medical Device Items and Grading by Item*. MFDS Notification No. 2025-95. Appendix: In Vitro Diagnostic Medical Device Items and Grading by Item. Cheongju, Republic of Korea.

[3] Ministry of Food and Drug Safety (MFDS) (2026). *Regulations on Medical Device Approval, Reporting, and Review*. MFDS Notification No. 2026-6. Cheongju, Republic of Korea.

[4] Ministry of Food and Drug Safety (MFDS) (2026). *Regulations on Medical Device Approval, Reporting, and Review*. MFDS Notification No. 2026-6. Appendix 5: Criteria for Judging Equivalent Products (related to Article 23). Cheongju, Republic of Korea.

[5] Ministry of Food and Drug Safety (MFDS) (2026). *Regulations on Medical Device Approval, Reporting, and Review*. MFDS Notification No. 2026-6. Appendix 7: Scope of Submitted Data Such as Technical Documents (related to Article 28). Cheongju, Republic of Korea.

[6] Ministry of Food and Drug Safety (MFDS) (2026). *Regulations on Medical Device Approval, Reporting, and Review*. MFDS Notification No. 2026-6. Appendix 15: Data to Be Submitted by Stage (related to Article 51(1)). Cheongju, Republic of Korea.

AI Co-Scientist Challenge Korea Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: **The abstract and introduction clearly state the scope and contributions of the paper, namely the proposal of a regulatory feasibility scoring framework based on intended use clarity and item classification, and its application to representative medical device and in vitro diagnostic use cases. The claims are explicitly framed as methodological and decision-support contributions rather than empirical performance improvements. Aspirational goals, such as supporting resource-rational regulatory strategies, are presented as motivation rather than achieved outcomes. All claims are consistent with the analytical case comparisons and regulatory interpretations provided in the paper, and no claims extend beyond the demonstrated scope.**

   Guidelines:

   - The answer NA means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A No or NA answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

Justification: **The paper includes a dedicated "Limitations and Future Work" section that explicitly discusses the scope and constraints of the proposed framework. It acknowledges reliance on current regulatory classification tables, the absence of empirical validation or automation, and the exclusion of post-market, reimbursement, and cross-jurisdictional considerations. The authors also discuss how regulatory changes or context-specific interpretations may affect generalizability, and outline directions for extending the framework using dynamic updates and domain-specific AI tools.**

Guidelines:

- The answer NA means that the paper has no limitation while the answer No means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory Assumptions and Proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: **The paper does not present formal theoretical results, theorems, or proofs. Its contributions are conceptual and methodological, focusing on regulatory interpretation, classification logic, and structured feasibility assessment rather than mathematical theory.**

   Guidelines:

   - The answer NA means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental Result Reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [N/A]

   Justification: **The paper does not include experimental results or empirical evaluations. Instead, it relies on regulatory documents, publicly available classification tables, and structured case analysis. Therefore, experimental reproducibility requirements are not applicable.**

   Guidelines:

   - The answer NA means that the paper does not include experiments.

- If the paper includes experiments, a No answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While AI Co-Scientist Challenge Korea does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: : [N/A]

   Justification: **The paper does not involve datasets, models, or executable code. All sources used are publicly available regulatory documents issued by the Ministry of Food and Drug Safety, which are cited in the references. As no experiments or computational models are presented, code or data release is not applicable.**

   Guidelines:

   - The answer NA means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (https://nips.cc/public/guides/CodeSubmissionPolicy) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so "No" is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (https://nips.cc/public/guides/CodeSubmissionPolicy) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new

proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.

- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental Setting/Details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

   Answer : [N/A]

   Justification: **The paper does not involve model training, testing, or experimental pipelines. The analysis is based on regulatory classification criteria and structured qualitative scoring, which are fully described in the methodology section.**

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment Statistical Significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [N/A]

   Justification: **The paper does not include experimental results, statistical evaluations, or empirical measurements. Its contributions are methodological and conceptual, focusing on regulatory feasibility scoring derived from regulatory classification rules, publicly available regulatory documents, and structured case analysis. Therefore, statistical significance testing, error bars, or confidence intervals are not applicable to the claims made in this work.**

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - The authors should answer "Yes" if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g. negative error rates).
   - If error bars are reported in tables or plots, The authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments Compute Resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [N/A]

Justification: **The paper does not involve computational experiments, model training, simulations, or large-scale data processing. Consequently, there are no compute resources, execution times, or hardware requirements relevant to reproducing experimental results. The analysis is based on regulatory documents and qualitative scoring logic, which does not require specialized computational infrastructure.**

Guidelines:

- The answer NA means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.

- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code Of Ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics https://nips.cc/public/EthicsGuidelines?

   Answer: [N/A]

   Justification: **The research fully conforms to the NeurIPS Code of Ethics. The study is based exclusively on publicly available regulatory documents and publicly disclosed classification information. It does not involve human subjects, personal data, proprietary datasets, or sensitive information. No experiments, data collection, or model deployment were conducted. Anonymity has been preserved in accordance with submission requirements, and there are no conflicts of interest or ethical risks related to privacy, fairness, security, or misuse.**

   Guidelines:

   - The answer NA means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer No, they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader Impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: : [Yes]

    Justification: **The paper discusses both positive and potential negative societal impacts. On the positive side, the proposed regulatory feasibility scoring framework can improve resource-rational decision-making in medical device research, reduce late-stage project failure, and enhance alignment between public R&D investment and regulatory approval pathways. On the negative side, the paper acknowledges that misinterpretation or overreliance on feasibility scores could bias funding decisions against high-risk, long-term innovation. The discussion section emphasizes that the framework is intended as a decision-support tool rather than a deterministic approval mechanism and highlights the need for expert judgment to mitigate such risks.**

    Guidelines:

    - The answer NA means that there is no societal impact of the work performed.
    - If the authors answer NA or No, they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is

being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.

- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper does not release data, code, models, or other assets that pose risks of misuse or dual use. It introduces a conceptual framework and qualitative scoring methodology based on regulatory documents. Therefore, safeguards for responsible release are not applicable.

Guidelines:

- The answer NA means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [N/A]

    Justification: **The paper does not use external datasets, code, software packages, or trained models. All referenced materials are publicly available regulatory documents issued by a government authority and are cited appropriately in the references section. As no licensed research assets are reused or redistributed, licensing and terms-of-use disclosures are not applicable**

    .Guidelines:

    - The answer NA means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New Assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [N/A]

    Justification:

    The paper does not introduce or release new assets such as datasets, code, trained models, benchmarks, or other reusable research artifacts. All materials referenced in the study consist of publicly available regulatory documents and classification tables issued by the Ministry of Food and Drug Safety. As no new assets are created or distributed, asset documentation and release requirements are not applicable.

    Guidelines:

    - The answer NA means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license,

limitations, etc.

- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and Research with Human Subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: **The paper does not involve crowdsourcing, surveys, interviews, experiments with human participants, or any form of human subject research. The analysis is entirely document-based and conceptual, relying on publicly available regulatory information and hypothetical case analysis. Therefore, participant instructions, consent procedures, and compensation details are not applicable.**

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification**: This study does not involve human subjects, personal data, biological samples, or interactions with individuals. Consequently, no risks to participants exist, and Institutional Review Board (IRB) approval or equivalent ethical review was not required. The research complies fully with the NeurIPS Code of Ethics by avoiding human subject involvement altogether.**

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.